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Training-Free Geometric Rectification Framework for Extreme Few-Shot Satellite Image Classification

Lam Mai¹, Tan-Khoi Nguyen², Van-Dung Hoang³, Hoang-Huu-To Nguyen¹, Dinh-Nguyen Le¹

¹The University of Danang - Vietnam-Korea University of Information and Communication Technology

²The University of Danang - University of Science and Technology

³Ho Chi Minh City University of Technology and Engineering

Background & Problem

- Context: Satellite classification is vital for disaster response and emergency surveillance.
- The Extreme Few-Shot Challenge:
 - Labeled data is scarce in emergency scenarios.
 - Real constraint: Only 5 to 10 images per class (Micro-sets).
- Limitations of Deep Learning:
 - Requires fine-tuning and weight updates $\nabla\theta L$.
 - Causes Catastrophic Overfitting and Feature Collapse in low-data regimes.
 - Heavy computational and GPU time cost.



a) Flooded Area



b) Wildfire Scar



c) Earthquake Damage

Our Contributions

- Training-Free: Zero parameter updates, zero backpropagation cost.
- Non-Parametric Geometric Attention: Fuses global context & local patches to suppress background noise.
- Tukey Transform: Solves the Hubness problem, reshaping spiky distributions into Gaussian-like manifolds.

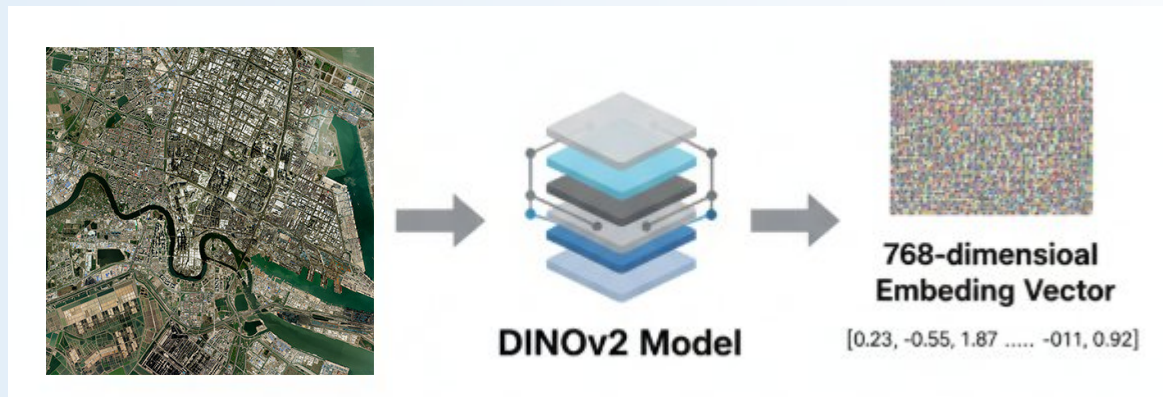


Fig 2: Frozen DINOv2 backbone extracting high-dimensional feature representations

Proposed TGRF Architecture

- Our TGRF pipeline - deterministic 4-step feed-forward process.
- Feed-forward processing only; fast, stable, and deterministic.

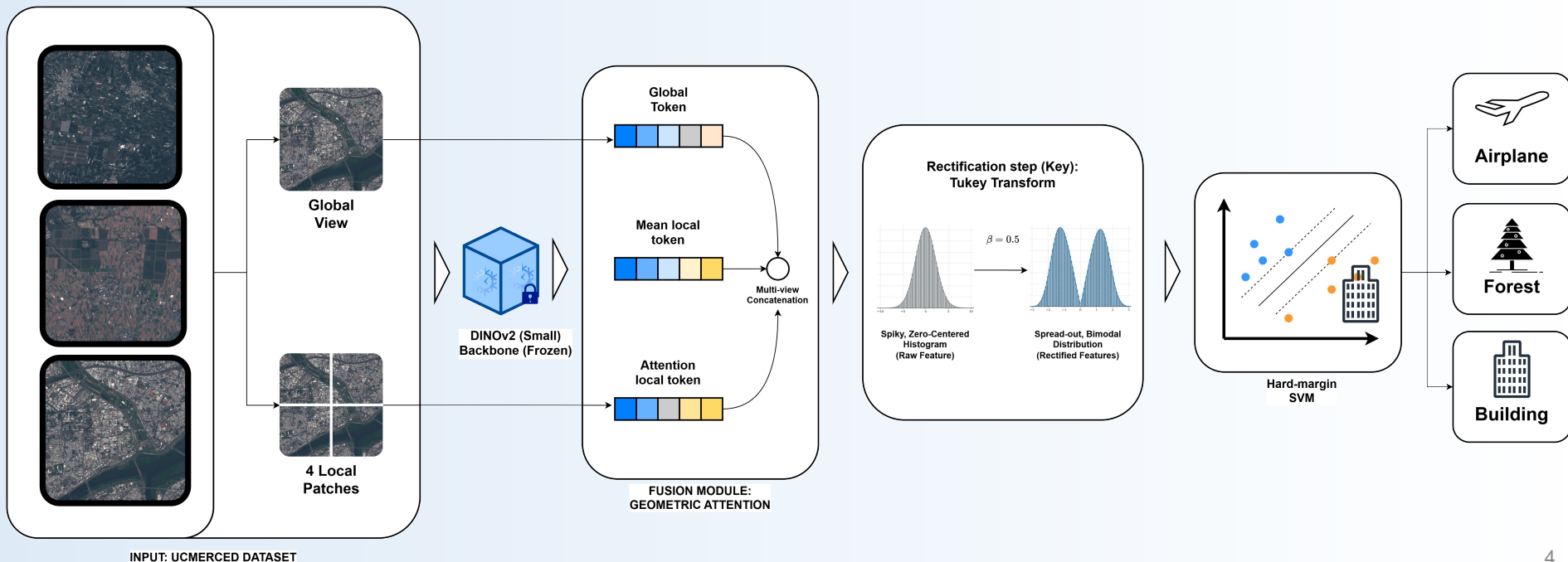


Fig 3: Proposed TGRF Architecture

Multi-View Extraction & Geometric Attention

- Global View

$$X_g(224^2) \rightarrow \mathbf{z}_g \in \mathbb{R}^{384}$$

- Local Views

$$X_l = \{p_1, p_2, p_3, p_4\} \text{ (quadrant crops)} \rightarrow \mathbf{Z}_l \in \mathbb{R}^{4 \times 384}$$

- Spherical Projection: Normalize features to unit sphere $\mathbb{S}^{D-1}$ to remove contrast variations.
- Winner-Take-All Attention ($\tau = 0.05$)

- Cosine similarity:

$$s_i = \langle \hat{\mathbf{z}}_g, \hat{p}_i \rangle$$

- Softmax scaling:

$$\alpha_i = \frac{\exp(s_i / \tau)}{\sum_j \exp(s_j / \tau)}$$

- Dynamically suppresses background noise, preserving discriminative foreground features.

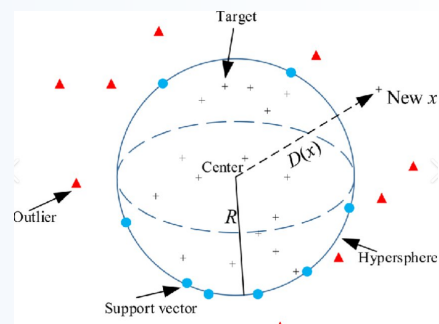


Fig 4: Hypersphere in high-dimensional space

Feature Rectification

- The Hubness Problem: Raw DINOv2 features have high kurtosis (Laplacian peak at $x = 0$), causing feature vectors to crowd near the origin.
- Tukey's Power Transform ($\beta = 0.5$)

$$\phi(x) = \text{sign}(x)|x|^\beta$$

- Derivative Analysis

$$\frac{d\phi(x)}{dx} = \frac{1}{2\sqrt{x}} \rightarrow \infty \quad \text{as } x \rightarrow 0$$

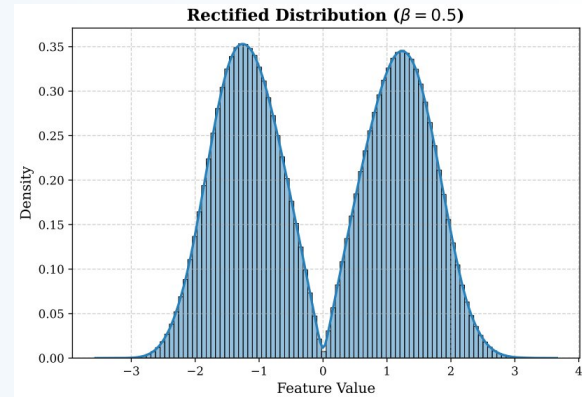
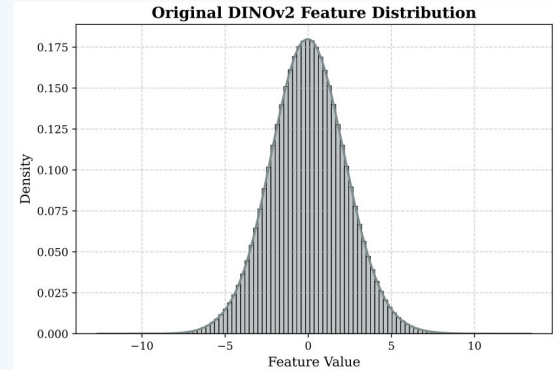


Fig 5: Feature distribution before (spiky Laplacian) and after Tukey transform (bimodal Gaussian-like).

Extreme Augmentation & Hard-Margin SVM

- 1-shot limitation: A single sample vector cannot represent the true class distribution.
- Synthetic Augmentation: Generate $N_{aug} = 20$ synthetic variations per support image using rotations, flips, and color jitter.
- Point-Cloud Separation: Converts a fragile point-to-point boundary into a robust cloud-to-cloud manifold boundary.
- Hard-Margin Linear SVM ($C = 50$):
Optimizes separating hyperplanes with maximum geometric margin.

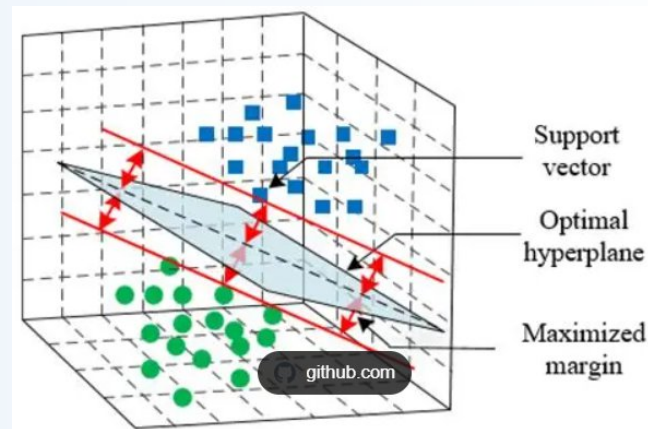


Fig 6: Support vector machine maximum margin hyperplane diagram

Experimental Setup

- **Dataset 1: UC Merced (UCM)**

- 2,100 images, 21 classes, 100 images/class, 0.3 m high resolution.
- Episodic 5-way K -shot protocol, $K \in \{1, 2, 5\}$.
- 100 episodes $\times$ 3 seeds.

- **Dataset 2: Sentinel-2 Micro-Set**

- Strictly 50 images total, comprising 5 classes: City, Coastal, Farmland, Mixed, Mountains.
- 10 images/class.
- 5-Fold Stratified Cross-Validation for rigorous evaluation.



Fig 7: Samples of UC Merced dataset (0.3m resolution) showing high intra-class diversity

Quantitative Results

Table 1. Main Results on UCM Dataset (Augmented). Comparison of mean accuracy (%) $\pm$ standard deviation over 3 seeds. TGRF achieves state-of-the-art performance.

Shot (k)	ProtoNet	k-NN	LogReg	TGRF (Ours)
1-Shot	80.43 $\pm$ 1.45	81.72 $\pm$ 1.52	79.07 $\pm$ 1.32	81.73 $\pm$ 0.84
2-Shot	87.68 $\pm$ 0.53	88.15 $\pm$ 0.62	87.72 $\pm$ 1.10	89.87 $\pm$ 0.88
5-Shot	92.09 $\pm$ 0.98	92.64 $\pm$ 0.69	94.32 $\pm$ 0.53	95.04 $\pm$ 0.57

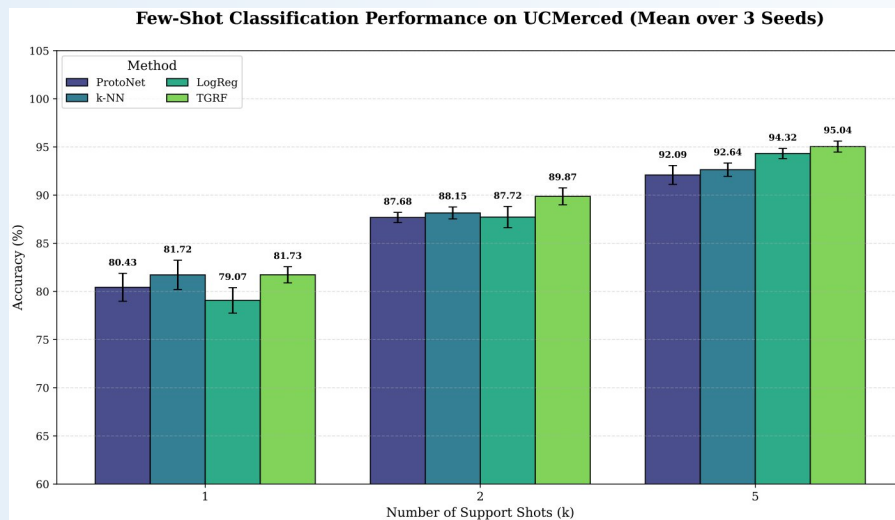


Fig 8: Classification mean accuracy over 3 seeds on UC Merced

Ablation Study

Table 3. Ablation Study: UCM Results WITHOUT Augmentation. Note the performance drop compared to Table 1.

Shot (k)	ProtoNet	k-NN	LogReg	TGRF (No Aug)
1-Shot	80.99 ± 0.51	31.61 ± 0.78	81.64 ± 0.53	79.40 ± 0.71
2-Shot	88.92 ± 0.28	80.92 ± 0.22	90.18 ± 0.46	85.01 ± 0.41
5-Shot	93.72 ± 0.87	91.08 ± 1.05	95.16 ± 0.60	93.28 ± 0.65

Table 4. Ablation Study: Sentinel-2 Results WITHOUT Augmentation.

Metric	ProtoNet	k-NN	LogReg	TGRF (No Aug)
Mean Acc	77.33%	77.33%	83.33%	81.33%
Std Dev	± 13.35	± 13.35	± 11.75	± 13.02

Qualitative Visualizations

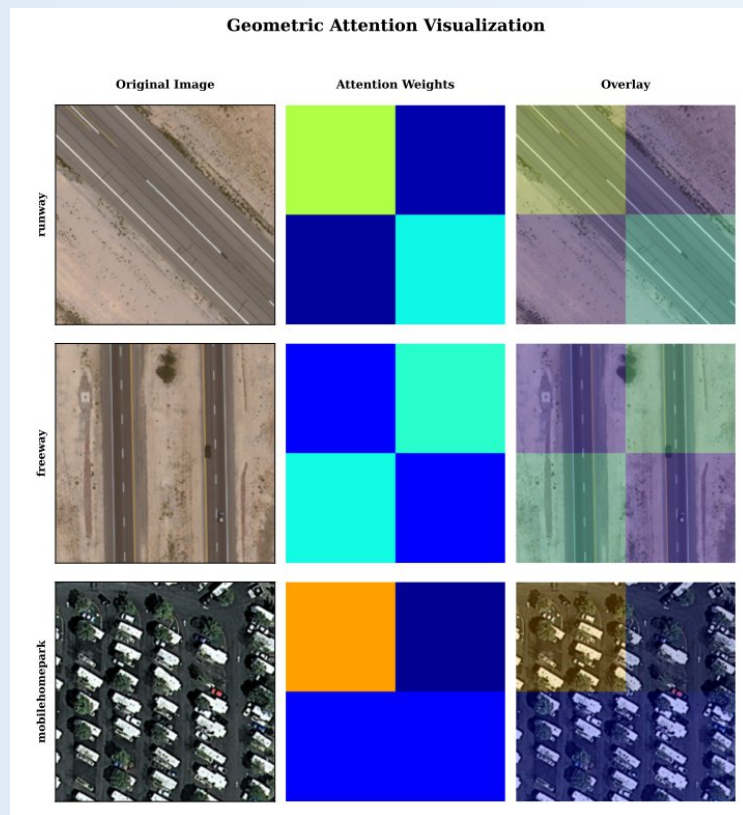


Fig 9: Autonomous semantic localization without trainable parameters

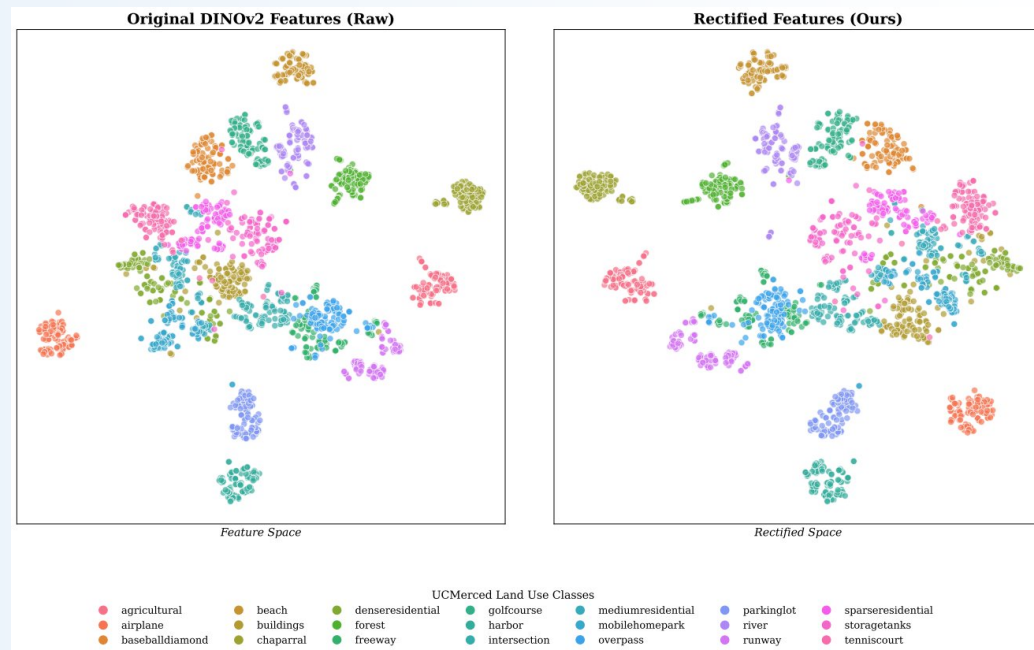


Fig 10: t-SNE manifold: Raw entangled features (left), TGRF rectified clean clusters (right)

Conclusion



- Geometric calibration of pre-trained feature manifolds outperforms gradient fine-tuning in extreme low-data regimes.
- TGRF is training-free, fast, and achieves state-of-the-art accuracy.
- Extension to multispectral (SAR/Radar) imagery and integration with multimodal foundation models.

References / Trích dẫn

Thank you
for your attention